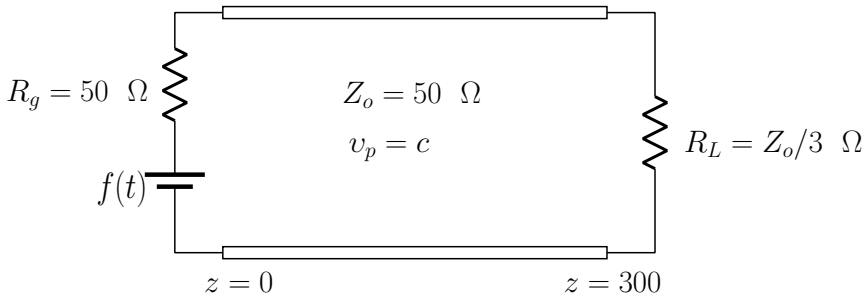
1. **Bounce diagram problem :**
The associated circuit is shown in the following figure:



a) (2.5 pts.) In this circuit, the **injection coefficient** is found as

$$\tau_g = \frac{Z_o}{R_g + Z_o} = \frac{1}{2}.$$

The load voltage reflection coefficient is given by

$$\Gamma_{L_V} = \frac{R_L - Z_o}{R_L + Z_o} = -\frac{1}{2},$$

whereas the generator voltage reflection coefficient is

$$\Gamma_{g_V} = \frac{R_g - Z_o}{R_g + Z_o} = 0.$$

The corresponding current reflection coefficients are

$$\Gamma_{L_I} = -\Gamma_{L_V} = \frac{1}{2}, \text{ and}$$

$$\Gamma_{g_I} = -\Gamma_{g_V} = 0.$$

Using these information, we can build the following "bounce diagrams" for the voltage $V(z,t)$ and current $I(z,t)$ on the transmission line.

b) (2.5 pts.) Referring to the full expressions of $V(z,t)$ and $I(z,t)$ given in page 7 of Lecture 21, we can write

$$V(z,t) = \frac{1}{2}\delta(t - \frac{z}{c}) - \frac{1}{4}\delta(t + \frac{z}{c} - 2\mu\text{s})$$
$$I(z,t) = \frac{1}{100}\delta(t - \frac{z}{c}) + \frac{1}{200}\delta(t + \frac{z}{c} - 2\mu\text{s}).$$

Evaluating these expressions at $z = \frac{l}{2} = 150\,\text{m}$, we have

$$V(\frac{l}{2},t) = \frac{1}{2}\delta(t - 0.5\mu\text{s}) - \frac{1}{4}\delta(t - 1.5\mu\text{s})$$
$$I(\frac{l}{2},t) = \frac{1}{100}\delta(t - 0.5\mu\text{s}) + \frac{1}{200}\delta(t - 1.5\mu\text{s}).$$

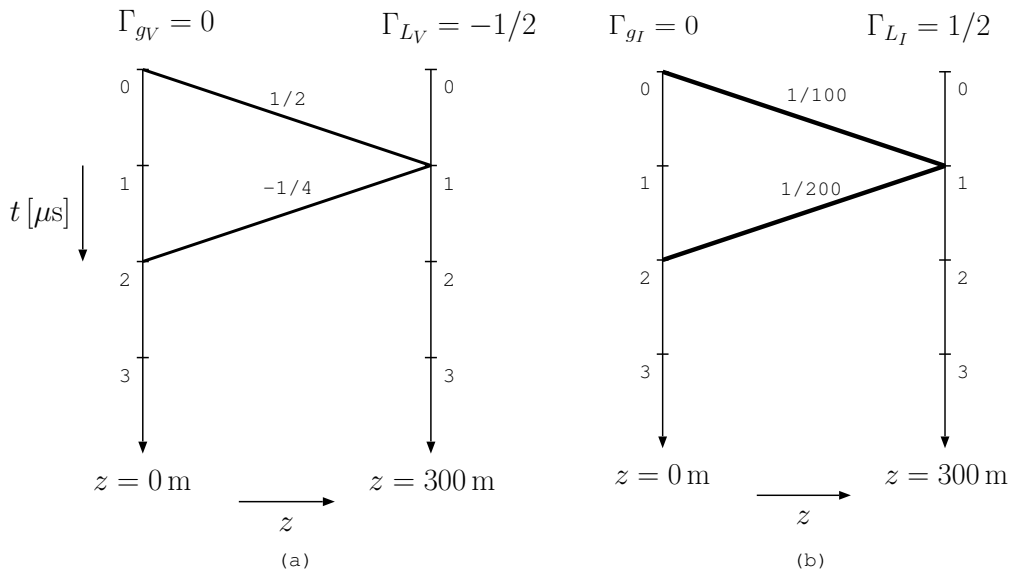


Figure 1: (a) Voltage and (b) current bounce diagrams for voltage source $f(t) = \delta(t)$.

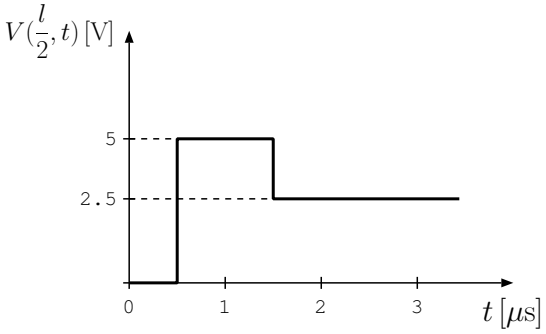
c) (2.5 pts.) Considering the circuit in the problem as a system, its impulse voltage response at $z = \frac{\ell}{2}$ can be written as

$$V(z,t) = h_z(t)\Big|_{z=\frac{\ell}{2}} = \frac{1}{2}\delta(t - 0.5\mu\text{s}) - \frac{1}{4}\delta(t - 1.5\mu\text{s}).$$

Given that $f(t) = 10u(t)\,\text{V}$, we can compute $V(\frac{l}{2}, t)$ as follows

$$V(\frac{l}{2}, t) = f(t) * h_z(t) = 5u(t - 0.5\mu\text{s}) - 2.5u(t - 1.5\mu\text{s})\,\text{V}.$$

The diagram of $V(\frac{\ell}{2}, t)$ is shown in the following figure.



d) (2.5 pts.) Given that the voltage source is $f(t) = 10u(t)$, $V(z,t)$ and $I(z,t)$ can also be written as

$$\begin{aligned} V(z,t) &= 5u(t - \frac{z}{c}) - 2.5u(t + \frac{z}{c} - 2\mu\text{s})\,\text{V}, \\ I(z,t) &= \frac{1}{10}u(t - \frac{z}{c}) + \frac{1}{20}u(t + \frac{z}{c} - 2\mu\text{s})\,\text{A}. \end{aligned}$$

By plotting the $V(z,t)$ and $I(z,t)$ as a function of time, we can find the steady state voltage and current as

$$V(z,t) = 2.5\,\mathrm{V}, \quad \text{and} \quad I(z,t) = \frac{3}{20}\,\mathrm{A},$$

respectively.

2. **Bonus problem :**

a) (5 pts.) The voltage drop between the plates is given by $V = \int \mathbf{E}.d\ell$. Hence, we write

$$\begin{aligned} V(z,t) &= \int_{x=0}^{d} \mathbf{E}(z,t)\cdot\hat{x}dx \\ &= \int_{x=0}^{d} 4\cos(\omega(t-\frac{z}{c}))dx \\ &= 4d\cos(\omega(t-\frac{z}{c})), \end{aligned}$$

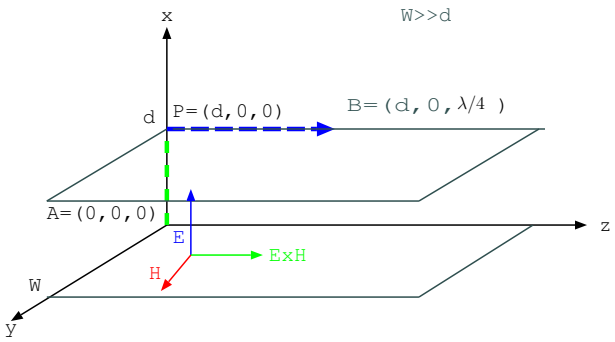
where $d = 0.01\,\mathrm{m}$. Thus, we write

$$V(0,t) = 4\times10^{-2}\cos(\omega t)\,\mathrm{V}.$$

b) (5 pts.) Yes, it is the same since the electric field has only x-polarized.

c)

i. (2.5 pts.) The path from A to P and then from P to B is shown in the folllowing figure along straight line segments:

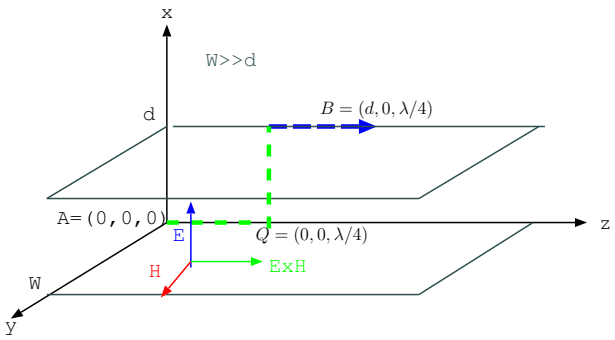


The line vector from A to B can be composed into two sections, i.e. $d\mathbf{l} = \hat{x}dx + \hat{y}dy + \hat{z}dz$ Therefore, the voltage drop from A to B is given by

$$V_{AB}(z,t) = \int_{A\to P} \mathbf{E}(z,t)\cdot d\mathbf{l} + \int_{P\to B} \mathbf{E}(z,t)\cdot d\mathbf{l} = \int_{0}^{d} \mathbf{E}(z,t)\cdot\hat{x}dx.$$

Finally, we get

$$V_{AB}(t) = 4\times10^{-2}\cos(\omega t)\,\mathrm{V}.$$



ii. (2.5 pts.) The path from A to Q and then from Q to B is shown in the folllowing figure along straight line segments: Following the variation along each axis in the figure, the voltage drop from A to B can be given by

$$V(z,t) = \int_{A\to Q} \mathbf{E}(z,t)\cdot d\mathbf{l} + \int_{Q\to B} \mathbf{E}(z,t)\cdot d\mathbf{l} = \int_0^d \mathbf{E}(z,t)\cdot \hat{x}dx$$

Since the wave traveled by a distance of $z = \frac{\lambda}{4}$, we write

$$V_{AB}(t) = 4d\cos(\omega t - \frac{2\pi}{\lambda}z) = 4\times 10^{-2}\sin(\omega t)\,\text{V}.$$

Since the electric field given in the problem is NOT curl-free (i.e. $\nabla\times\mathbf{E}\neq 0$), line integral will be path-dependent. This is why we see different functions obtained for $V_{AB}(t)$. For more information, please see Lecture 4 notes.

3.

a) (2 pts.) Looking at the voltage waveform plot given in the problem, it can be seen that the amplitude of the incident pulse at $t = 1\,\mu$s is 20 V. Applying the voltage division formula as we did in page 3 of Lecture 28, i.e. $f(t) = \tau_g f_i(t)$ where $f_i(t)$ is the generator voltage. Thus, we can write

$$20 = 60\times\frac{Z_o}{R_g+Z_o} \Longrightarrow Z_o = 30\,\Omega.$$

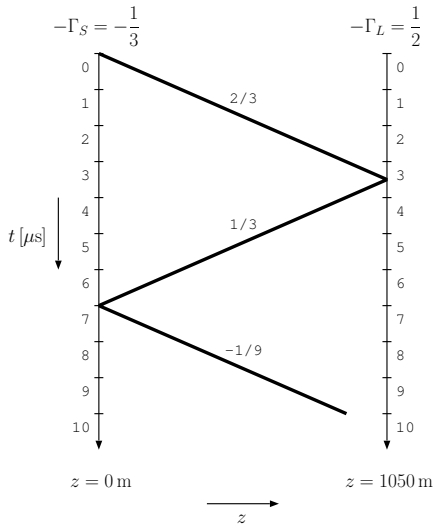
b) (2 pts.) Again looking at the plot in the problem, one can see that the amplitude of the first reflected pulse is -10 V. Therefore, the reflection coefficient is

$$\Gamma_L = -\frac{10}{20} = \frac{R_L - Z_o}{R_L + Z_o}$$

from which we obtain the load resistance as $R_L = \frac{Z_o}{3} = 10\,\Omega$.

c) (2 pts.) Since at $z = 300$ m, the incident pulse is delayed by $1\,\mu$s, the propagation speed $v_p = \frac{300}{1\times 10^{-6}} = 3\times 10^8$ m/s $\approx c$, where c is the speed of light. The time interval between the incident pulse and the second reflected pulse is $7\,\mu$s which is equal to the two-way travel time. Then, the time a pulse travels in one-way along the distance L is $\tau_L = 3.5\,\mu$s. Thus, we can find that the length of the line is $L = v_p\times\tau_L = 300\times 3.5 = 1050$ m.

d) (2 pts.) Bounce diagram for the *current* waveform $I(z,t)$ for $0 < t < 10\,\mu$s is given below:



e) (2 pts.) The expression for the current waveform for $0 < z < L$ and $0 < t < 10\,\mu$s is

$$I(z,t) = \frac{2}{3}\delta(t - \frac{z}{v_p}) + \frac{1}{3}\delta(t + \frac{z}{v_p} - 7\,\mu\text{s}) - \frac{1}{9}\delta(t - \frac{z}{v_p} - 7\,\mu\text{s})\,\text{A}.$$

4. a) (2.5 pts.) Considering that the source voltage is given by $f(t) = u(t)$, it is directly seen from the plot that the injection coefficient $\tau_g = 0.5$. Hence, utilizing $\tau_g = \frac{Z_o}{R_g + Z_o}$, the characteristic impedance of the T.L. is found to be $Z_o = R_g = 100\,\Omega$.

b) (2.5 pts.) The plot has a jump at $t = 2\,$ns, and we can infer that it takes the wave 2 ns to start from the source, then get reflected by the defect, finally go back to the source. Based on this, we have

$$\frac{2d}{v} = 2\,\text{ns}.$$

Therefore, $d = 0.2\,$m.

c) (2.5 pts.) Again, from the plot we know that

$$\begin{aligned} \tau_g \Gamma_d &= 1 - 0.5 \\ \frac{1}{2}\Gamma_d &= 0.5 \end{aligned}$$

from which we get $\Gamma_d = 1$. On the other hand, we have

$$\Gamma_d = \frac{(R_d + Z_o) - Z_o}{(R_d + Z_o) + Z_o} = \frac{R_d}{R_d + 2Z_o},$$

where R_d denotes the effective series resistance of the defect. We see R_d should be very/infinitely large $R_d \to \infty$. So the defect is actually an open circuit.

d) (2.5 pts.) Since all the incident wave get reflected at the defect as $\Gamma_d = 1$, there will be no voltage response at the load.

5. (Bonus problem) (20pts) This problem is similar to that given in Example 3 in Lecture 30.

a) (5 pts)The voltage in the first line will be a sum of incident $V^+$and reflected voltages V^-, i.e. $V_1 = V^+ + V^-$ whereas in the second line it will be $V_2 = V_R + V^{++}$ where V_R is the voltage across the resistor. Therefore, the KVL equation at the junction is

$$V_1 = V_2 \implies V^+ + V^- = V_R + V^{++}.$$

The current flowing in the first line is the sum of the incident and reflected currents, i.e. $I_1 = \frac{V^+}{Z_1} - \frac{V^-}{Z_1}$. However, this current must also be equal to the current flowing through the resistor and also to the current transmitted to the second line. Thus, the KCL equation at the junction is given by

$$\frac{V^+ - V^-}{Z_1} = \frac{V_R}{R} = \frac{V^{++}}{Z_2}.$$

b) (5 pts)Combining the previous equations by eliminating $V^{++}$and V_R we have

$$\frac{V^+}{Z_1} - \frac{V^-}{Z_1} = \frac{V^+ + V^-}{R + Z_2} \quad \rightarrow \quad \left(\frac{1}{Z_1} - \frac{1}{Z_{eq}}\right) V^+ = \left(\frac{1}{Z_1} + \frac{1}{Z_{eq}}\right) V^-,$$

where $Z_{eq} = R + Z_2$. Thus, the reflection coefficient is

$$\Gamma_{12} = \frac{V^-}{V^+} = \frac{Z_{eq} - Z_1}{Z_{eq} + Z_1},$$

Similarly, transmission coefficient is simply found by referring to $\tau_{12} = \frac{V^{++}}{V^+}$. Hence, we have

$$\begin{aligned} \tau_{12} &= (1 + \Gamma_{12}) \frac{Z_2}{R + Z_2} \\ &= \left(1 + \frac{Z_{eq} - Z_1}{Z_{eq} + Z_1}\right) \frac{Z_2}{Z_{eq}} \\ &= \frac{2Z_2}{Z_{eq} + Z_1}, \end{aligned}$$

where $\frac{Z_2}{R+Z_2}$ is a voltage division factor and $Z_{eq} = R + Z_2$.

c) (5 pts)Considering $Z_1 = 100\,\Omega$, $Z_2 = 50\,\Omega$, and $R = 25\,\Omega$, we can find that $Z_{eq} = 75\,\Omega$. Then, we calculate

$$\Gamma_{12} = \frac{Z_{eq} - Z_1}{Z_{eq} + Z_1} = \frac{75 - 100}{75 + 100} = -\frac{1}{7},$$

and

$$\tau_{12} = \frac{2Z_2}{R + Z_1 + Z_2} = \frac{2 \times 100}{25 + 100 + 50} = \frac{4}{7}.$$

Note that in this case $1 + \Gamma_{12} \neq \tau_{12}$.

d) (5 pts)As the generator has no internal resistance, $\tau_g = 1$, and $\Gamma_g = -1$. The transit time across the transmission line is $t_d = \frac{\ell}{v} = \frac{2 \times 300\,\text{m}}{300\,\text{m}/\mu\text{s}} = 2\,\mu\text{s}$. Also, the reflection coefficient at the load side will be $\Gamma_L = 1$ as Z_2-line is open circuit terminated. The bounce diagram based on these information is given in Figure 2

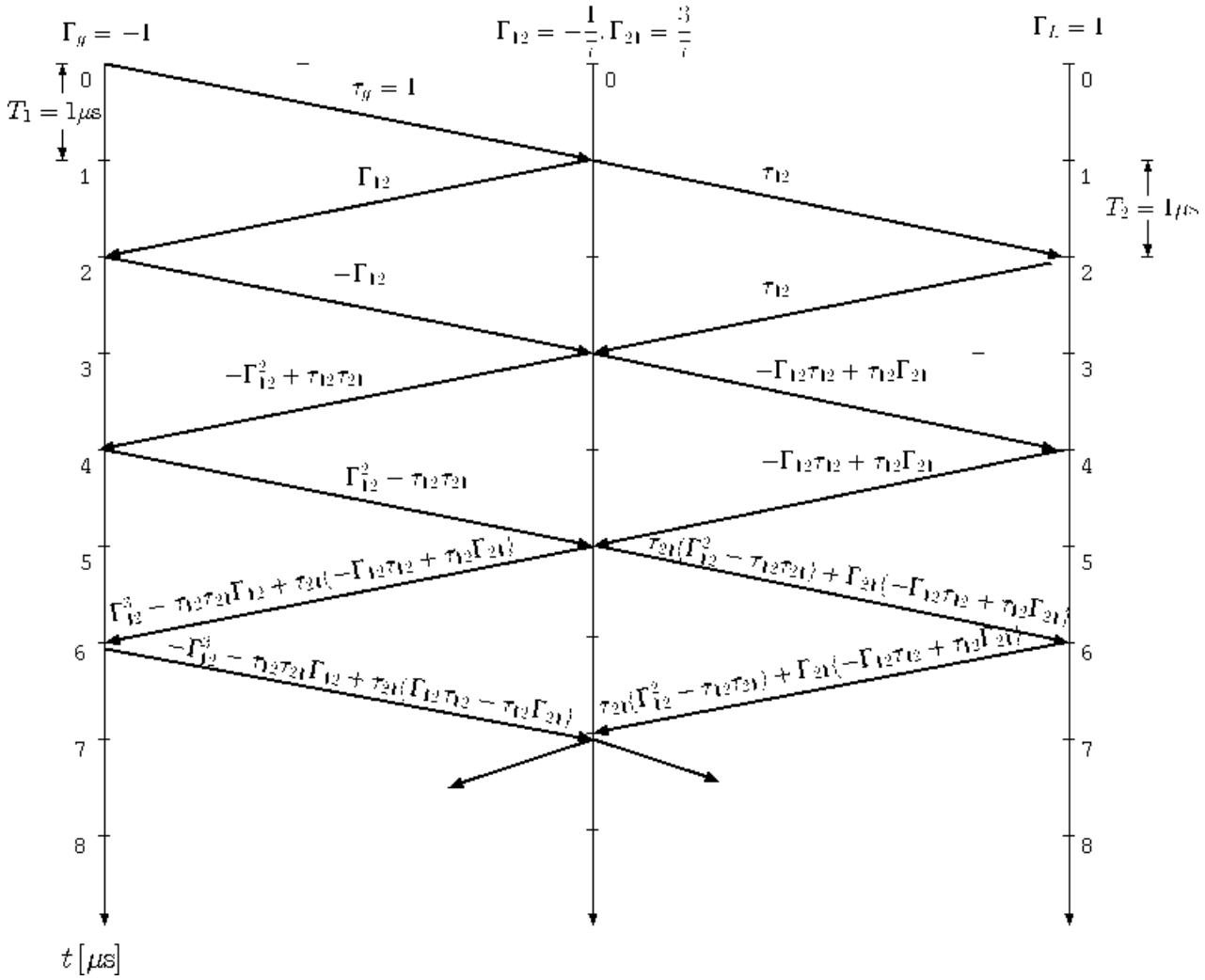


Figure 2: Bounce diagram for problem 5.